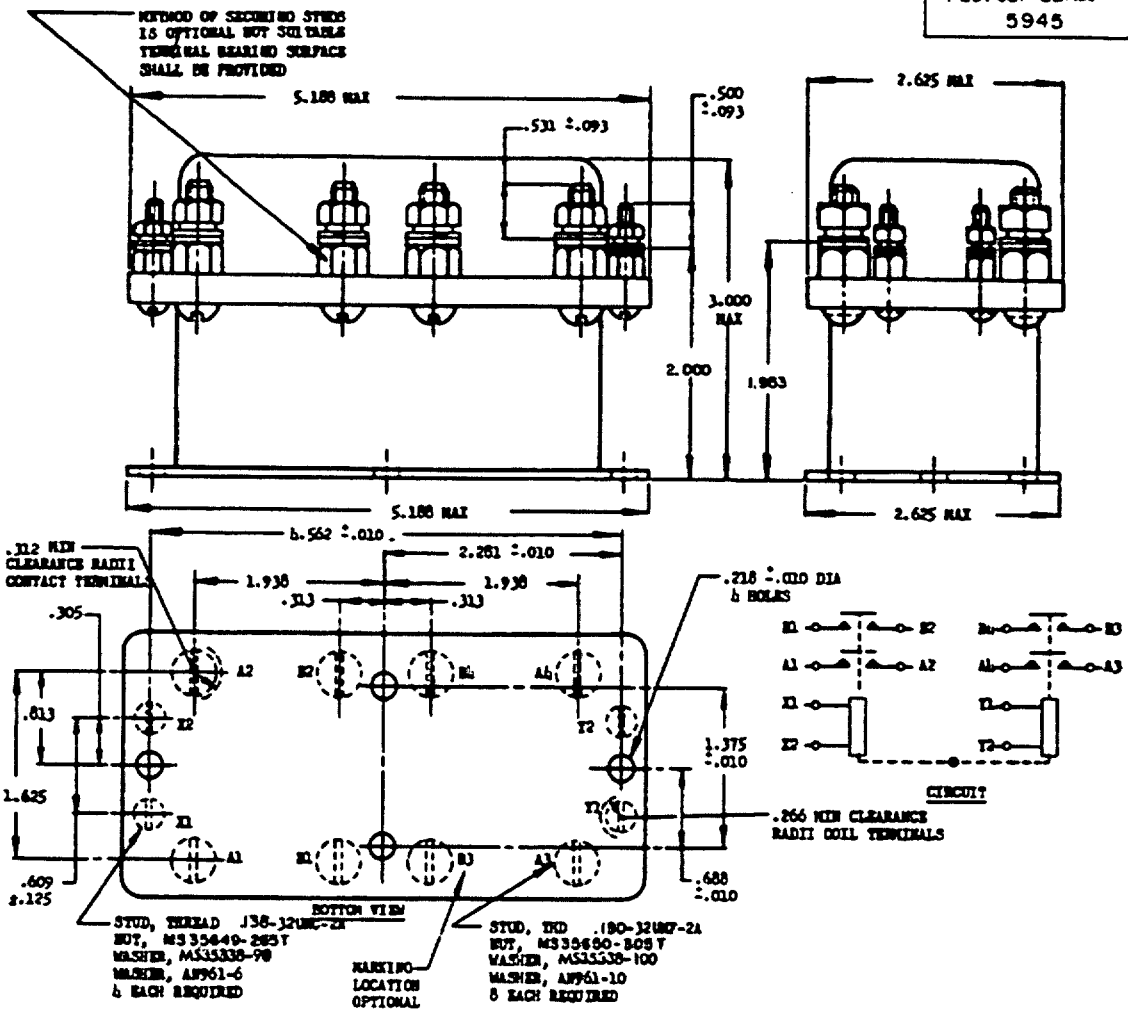
FED. SUP CLASS
5945



NOTES:-

1) DIMENSIONS ARE IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES - DECIMALS ± .062
2) ADDITIONAL FLAT WASHER MAY BE USED FOR TERMINAL SEAT
3) MECHANICAL LINKAGE SHALL BE PROVIDED TO PREVENT BOTH RELAYS BEING CLOSED AT THE SAME TIME
4) PART NUMBERS MS25030-2 AND -3 INACTIVE FOR DESIGN AFTER 25 NOVEMBER 1957
PART NUMBERS MS25030DIA AND DIB REPLACE -2 AND -3 RESPECTIVELY
PART NUMBER MS25030 DIA CANCELED AFTER 30 DECEMBER 1966. USE MS25030 DIB
5) FOR DESIGN FEATURE PURPOSES, THIS STANDARD TAKES PRECEDENCE OVER PROCUREMENT DOCUMENTS REFERENCED HEREIN
6) REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATION FOR BID

(L) DENOTES CHANGE

MS PART NUMBER	TYPE	COIL TYPE	TERMINAL TYPE	MOUNTING OR MATING SOCKET	AUXILIARY CONTACTS	MAX WEIGHT IN POUNDS
MS25030-0 1 B	II	DC	STUD	PLATE	N/A	1.7

P.A AF-85
Other Cust NAVY-AS

TITLE
RELAY, 50 AMP, 2PDT N O TYPE II, NON-HERMETICALLY SEALED, MECHANICALLY INTERLOCKED

MILITARY STANDARD
MS25030

Procurement Specification
MIL-R-6106

SUPERSEDES:

PAGE 1 OF 3

DD FORM 1 SEP 63 672 (Coordinated) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE

PROJECT 5945-0607-8

APPROVED 23 MARCH 1953 REVISED (H) 19 FEB 19(J) 1 APR 81 (K) 20 Nov 81 (L) 31 AUG 82

User activities: Army, Navy, Air Force
Reviewer activities: USAF 99, 11 Navy - EC

This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document when applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T15:01Z
Check the source to verify that this is the current version before use.

FED. SUP CLASS
5945

OPERATING CHARACTERISTICS

MS part no.	Coil	Coil Data: Nominal Volts	Nominal Freq Hz	Nominal Ω Res +15% -10%	Max Volts	Max Amp	Max pick-up voltage 1/: Normal 1	Max pick-up voltage 1/: High temp test	Max pick-up voltage 1/: Cont current test	Drop out voltage 1/	Hold voltage 1/	Time milliseconds - Max 2/: Operate	Time milliseconds - Max 2/: Release	Time milliseconds - Max 2/: Bounce Main NO	Bounce Main NC	Bounce Aux NO	Bounce Aux NC
MS25030-018	X1,X2 Y1,Y2	28	DC	66	29	.5	18	21	22.5	1.5	7.0	25	15	5.0	--	--	--

RATED CONTACT LOAD (AMPERES PER POLE) CASE GROUNDED

TYPE OF LOAD	LIFE OPERATING CYCLES X 10 3	28 VDC Main NO	28 VDC Main NC	28 VDC Aux NO	28 VDC Aux NC	115 VAC, 1 PHASE Main 400 Hz	115 VAC, 1 PHASE Main 60 Hz	115 VAC, 1 PHASE Aux 400 Hz	115 VAC, 1 PHASE Aux 60 Hz	115/200 VAC, 3 PHASE 3/ Main 400 Hz	115/200 VAC, 3 PHASE 3/ Main 60 Hz	115/200 VAC, 3 PHASE 3/ Aux 400 Hz	115/200 VAC, 3 PHASE 3/ Aux 60 Hz	See Appropriate Notes
RESISTIVE	50	50	-	-	-	50	-	-	-	-	-	-	-	
INDUCTIVE	10	50	-	-	-	-	-	-	-	-	-	-	-	
INDUCTIVE														
MOTOR	50	50	-	-	-	35	-	-	-	-	-	-	-	
LAMP														
TRANSFER, LOAD														4/
MECHANICAL LIFE REDUCED CURRENT	100	12.5	-	-	-	12.5	-	-	-	-	-	-	-	
MIN CURRENT	50	5	-	-	-	-	-	-	-	-	-	-	-	

ENVIRONMENTAL CHARACTERISTICS

TEMPERATURE RANGE	-55 TO +71 °C
MAX ALTITUDE RATING	50,000 FT
SHOCK G-LEVEL	25 G
DURATION	6-9 mS
MAX DURATION CONTACT OPENING	2 mS
VIBRATION - SINUSOIDAL (SEE CHART BELOW) G-LEVEL	G
FREQUENCY RANGE	Hz
VIBRATION-RANDOM APPLICABLE SPEC	N/A
POWER SPECTRAL DENSITY RMS G MIN	N/A
FREQUENCY RANGE CURVE	N/A
HIGH SHOCK	N/A
ACCELERATION	10 G

ELECTRICAL CHARACTERISTICS

MIN. INSULATION RESISTANCE, INITIAL	100 MEGOHMS
AFTER LIFE OR ENVIRONMENTAL TESTS	50 MEGOHMS

DIELECTRIC STRENGTH (SEA LEVEL) 2-5 SECONDS

	Initial 28 VDC	Initial 115 VAC	After Life Tests 28V	After Life Tests 115V
COIL TO CASE	1250	N/A	1000	N/A
AUX CONTACTS	1250	N/A	1000	N/A
ALL OTHER POINTS	1250	1500	1000	1125

DIELECTRIC STRENGTH (ALTITUDE) 1 MINUTE

	28 VDC	115 VAC
COIL TO CASE	500	N/A
AUX CONTACTS	500	N/A
ALL OTHER POINTS	500	500

MAX CONTACT DROP INITIAL	.150 VOLTS
AFTER LIFE TEST	.175 VOLTS
OVERLOAD CURRENT (N.O.)	400 AMPS
RUPTURE CURRENT (N.O.)	500 AMPS
DUTY RATING	CONTINUOUS
RFI SPEC (APPLICABLE TO COIL CIRCUITS OF AC OPERATED RELAYS)	

NOTES, APPLICATION

1/ Over temperature range.
2/ With nominal coil voltage.
3/ Absence of value indicates relay is not rated for 3 phase applications.
4/ Transfer load indicates relay suitable for transfer between unsynchronized AC power supplies at rating indicated.

DASH NO.	VIBRATION LEVEL 5-10 Hz	10-55 Hz	55-250 Hz	250-500 Hz	500-1500 Hz
018	.08 DA	.06 DA	2 G's	2 G's	

APPROVED 23 MARCH 1953 REVISED (L) FOR CHANGES SEE PAGE 3

User activities: Army, Navy, Air Force

Reviewer activities: Army, Navy, Air Force EC 99,11

This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

P.A AF-85
Other Cust NAVY-AS

TITLE: RELAY, 50 AMP, 2PDT NO TYPE II, NON-HERMETICALLY SEALED, MECHANICALLY INTERLOCKED

MILITARY STANDARD
MS25030

Procurement Specification MIL-R-6106

SUPERSEDES:

PAGE 2 OF 3

DD FORM 1 SEP 63 672 (Coordinated) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T15:01Z
Check the source to verify that this is the current version before use.

FED. SUP CLASS
5945

User activities: Army, Navy, Air Force

Reviewer activities: Army, Navy, Air Force

EC 99,11

APPROVED 23 MARCH 1953 REVISED Ⓛ FOR CHANGES SEE PAGE 3

Ⓛ Qualification By Similarity.

Part Number	Loads						Dynamics 1/			Environmental		
	Type 1			Type 1 ER								
	A	B	C	D	E	F	A	X	Y	A	B	C
MS25030-D 1B	4						2			4		

1/ All units must be tested, reference amendment 1, appendix 1, MIL-R-6106, construction, internal.

P.A AF-85 Other Cust NAVY-AS	TITLE RELAY, 50 AMP, 2PDT NO TYPE II, NON-HERMETICALLY SEALED, MECHANICALLY INTERLOCKED	MILITARY STANDARD MS25030
Procurement Specification MIL-R-6106	SUPERSEDES:	PAGE 3 OF 3

DD FORM 1 SEP 63 672 (Coordinated) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE

This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T15:01Z
Check the source to verify that this is the current version before use.